

Manuscript Review Report

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Important Notes

Version 1.0 (Beta): This tool was developed in a short timeframe and may contain hallucinations and errors. Approximately 50% of feedback will likely be unusable, 30% may be mediocre, and 20% should be helpful in highlighting issues that might have been overlooked. Please send feedback to rjakob@ethz.ch so we can continue to improve our agents.

Model Limitations: This report was generated using a basic model (ChatGPT 3.5). For more sophisticated analysis, the code is open access and users can run more advanced models (such as GPT-4) with their own API keys. Visit <https://github.com/robertjakob/rigorous> for more information.

Development Status: This tool is still in testing mode with many agents in their first iteration. We welcome contributions to build the best review agent team to improve the quality of scientific publishing. Join us at <https://github.com/robertjakob/rigorous>.

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Overall Assessment

The manuscript received an average score of 6.56/10 across all evaluation criteria. A total of 75 critical remarks and 124 improvement suggestions were identified.

Agent Reports

Section-Specific Agents (S1-S10)

S1 - Title and Keywords

Score: 7/10

Summary

The title effectively conveys the comparison between different front-of-package food labels, with a focus on the Spectrum Food Labels and FDA's Nutrition Info Label. The title could be further optimized for search engines to enhance discoverability. Overall, the title aligns well with field conventions and effectively communicates the study's focus to the audience.

Critical Remarks

No critical remarks identified.

Improvement Suggestions

Suggestion 1

Location: Title

Category: title

Focus: comprehensive_improvement

Original Text:

Testing FDA's proposed "Nutrition Info" label against other front-of-package food labels: A randomized trial

Improved Version:

Comparing Spectrum Food Labels to FDA's Nutrition Info Label: A Randomized Trial

Explanation:

The improved title maintains clarity by specifying the comparison between Spectrum Food Labels and FDA's Nutrition Info Label. It accurately represents the content of the study by highlighting the key labels being tested. The title captures attention by emphasizing the innovative Spectrum Food Labels and their comparison to the FDA's proposed label. It is optimized for search engines by including relevant terms like 'Spectrum Food Labels' and 'FDA's Nutrition Info Label'. The title follows field conventions by clearly indicating the experimental nature of the study.

Detailed Feedback

Title Analysis

The current title is clear and understandable, accurately representing the comparison between different front-of-package food labels. It captures attention by mentioning the FDA's proposed 'Nutrition Info' label. The title is somewhat optimized for search engines but could be improved by including more specific terms. It follows field conventions by indicating the experimental nature of the study.

Keywords Analysis

No keywords section found

Guidelines Compliance

The title and keywords section comply with field conventions by clearly indicating the experimental nature of the study and the comparison between different food labels.

Discoverability Assessment

The title could be further optimized for search engines by including specific terms related to the study, such as 'Spectrum Food Labels' and 'FDA's Nutrition Info Label'. This optimization would enhance the discoverability of the study.

Audience Alignment

The title effectively communicates the focus of the study to the audience, highlighting the comparison between different front-of-package food labels and the experimental nature of the research.

S2 - Abstract

Score: 7/10

Summary

The abstract demonstrates a solid structure and covers essential components of a research study. However, improvements are needed in providing more detailed methodology, contextualizing results, and strengthening the conclusion for better impact and clarity.

Critical Remarks

Remark 1

Category: clarity

Location: Abstract - Methods

Issue: The methodology description lacks details on randomization procedures, blinding, and potential biases.

Severity: medium

Impact: This omission undermines the credibility of the study and raises concerns about internal validity.

Remark 2

Category: content

Location: Abstract - Results

Issue: The presentation of results lacks context on the statistical significance and practical implications of the findings.

Severity: medium

Impact: Readers may struggle to interpret the significance of the results without this crucial information.

Remark 3

Category: standards

Location: Abstract - Conclusion

Issue: The conclusion lacks a clear summary of the study's implications and recommendations for future research or policy.

Severity: medium

Impact: A stronger conclusion is essential to provide closure and guide readers on the study's contributions.

Improvement Suggestions

Suggestion 1

Location: Abstract - Methods

Category: clarity

Focus: readability

Original Text:

We conducted a randomized trial in October-November 2024 (n=5,636 US adults).

Improved Version:

A randomized trial was conducted in October-November 2024 involving 5,636 US adults.

Explanation:

Improving sentence structure for better readability and clarity.

Suggestion 2

Location: Abstract - Results

Category: clarity

Focus: readability

Original Text:

The spectrum labels led participants to make healthier purchases compared to both the positive labels only...

Improved Version:

Spectrum labels resulted in healthier purchases compared to positive labels...

Explanation:

Simplify and streamline the sentence for clearer communication.

Suggestion 3

Location: Abstract - Conclusion

Category: content

Focus: impact

Original Text:

Together, these results suggest that adopting spectrum labels could improve diet quality without exacerbating health disparities.

Improved Version:

These findings indicate that implementing spectrum labels may enhance diet quality without worsening health disparities.

Explanation:

Enhancing the impact of the statement and improving clarity.

Detailed Feedback

Structure Analysis

The abstract follows a standard structure with clear sections like Objectives, Methods, Results, and Conclusions. However, the flow could be improved by providing more seamless transitions between sections.

Content Analysis

The abstract is mostly complete in terms of content, covering objectives, methods, results, and conclusions. However, some sections lack depth, such as the methodology description and the context for the results.

Clarity Assessment

The abstract is generally clear and concise, but there are instances of complex sentences and technical terms that may hinder readability for non-experts.

Standards Compliance

The abstract adheres to scientific writing standards in terms of structure and content presentation. However, it could benefit from more explicit references to prior literature and theoretical frameworks.

Impact Evaluation

The study addresses an important issue related to food labeling and consumer behavior. The results have implications for public health policies and consumer decision-making.

S3 - Introduction

Score: 7/10

Summary

The introduction provides a solid foundation by addressing the background context, problem statement, research objectives, significance of the study, and integrating relevant literature. However, improvements are needed to enhance the specificity of data, clarify comparisons, and strengthen the impact of the findings. Ensuring smoother transitions and providing more detailed explanations can enhance the overall quality and effectiveness of the introduction.

Critical Remarks

Remark 1

Category: significance

Location: Introduction, Paragraph 39

Issue: The significance of the problem is mentioned but lacks specific data or statistics to emphasize the severity of the issue.

Severity: medium

Impact: Weakens the justification for the research.

Remark 2

Category: structure

Location: Introduction, Paragraph 63

Issue: The transition to introducing the research gap is abrupt and could be smoother for better flow.

Severity: medium

Impact: Affects the coherence and organization of the introduction.

Improvement Suggestions

Suggestion 1

Location: Introduction, Paragraph 39

Category: significance

Focus: impact

Original Text:

Poor diet quality is responsible for approximately 500,000 deaths and more than \$50 billion in healthcare costs in the US every year.

Improved Version:

Poor diet quality is responsible for approximately 500,000 deaths and more than \$50 billion in healthcare costs in the US every year, highlighting the urgent need for effective interventions.

Explanation:

Adding 'highlighting the urgent need for effective interventions' provides a stronger call to action.

Suggestion 2

Location: Introduction, Paragraph 61

Category: problem

Focus: gap

Original Text:

It remains unknown, however, whether these other new labels outperform positive labels or Nutrition Info labels.

Improved Version:

However, it remains unclear whether these new labels outperform positive labels or Nutrition Info labels, creating a research gap.

Explanation:

Adding 'creating a research gap' explicitly states the gap in current knowledge.

Suggestion 3

Location: Introduction, Paragraph 64

Category: objectives

Focus: objectives

Original Text:

To address these gaps and inform policymaking, this study aimed to evaluate whether the Nutrition Info label and other new front-of-package labels lead to healthier purchases compared to positive labels and compared to one another.

Improved Version:

To address these gaps and inform policymaking, this study aimed to assess the impact of the Nutrition Info label and other new front-of-package labels on promoting healthier purchases compared to positive labels and each other.

Explanation:

Clarifying the study's aim and the comparison between labels enhances the objectives.

Suggestion 4

Location: Introduction, Paragraph 229

Category: significance

Focus: impact

Original Text:

Together, these results suggest that adopting spectrum labels could improve diet quality without exacerbating health disparities.

Improved Version:

These findings suggest that adopting spectrum labels could enhance diet quality while mitigating health disparities, presenting a significant opportunity for public health improvement.

Explanation:

Expanding on the potential impact of spectrum labels strengthens the significance.

Suggestion 5

Location: Introduction, Paragraph 241

Category: problem

Focus: gap

Original Text:

In this study, Nutrition Info labels similar to FDA's proposed labels did not lead to healthier selections than positive labels, regardless of whether they were presented alone or alongside positive labels.

Improved Version:

The study findings revealed that Nutrition Info labels, akin to FDA's proposed labels, failed to drive healthier selections compared to positive labels, irrespective of their presentation alongside positive labels.

Explanation:

Enhancing the clarity and specificity of the comparison strengthens the argument.

Suggestion 6

Location: Introduction, Paragraph 223

Category: significance

Focus: impact

Original Text:

In this study, spectrum food labels that rated foods from least to most healthy led participants to select healthier foods and beverages compared to positive, Nutrition Info, and high-in labels.

Improved Version:

The study demonstrated that spectrum food labels, rating foods from least to most healthy, significantly influenced participants to choose healthier food and beverages over positive, Nutrition Info, and high-in labels.

Explanation:

Adding 'significantly influenced' emphasizes the impact of spectrum labels.

Suggestion 7

Location: Introduction, Paragraph 227

Category: significance

Focus: impact

Original Text:

The spectrum labels led to overall healthier selections compared to all other labeling systems we tested.

Improved Version:

Overall, the spectrum labels consistently resulted in healthier selections compared to all other labeling systems examined.

Explanation:

Enhancing the clarity and consistency of the comparison strengthens the conclusion.

Suggestion 8

Location: Introduction, Paragraph 232

Category: significance

Focus: impact

Original Text:

One possible explanation for why the spectrum labels outperformed the positive and high-in labels is that the spectrum labels were applied to all products, rather than only to healthier or less healthy products.

Improved Version:

A potential reason for the superior performance of spectrum labels over positive and high-in labels is their application to all products, irrespective of their healthiness level.

Explanation:

Clarifying the reasoning behind the effectiveness of spectrum labels enhances the explanation.

Suggestion 9

Location: Introduction, Paragraph 239

Category: significance

Focus: impact

Original Text:

Our results align with prior studies finding that spectrum labels improve the healthfulness of food and beverage purchases and suggest that adopting spectrum labels could modestly improve diet quality.

Improved Version:

Consistent with previous research demonstrating the effectiveness of spectrum labels in enhancing the healthfulness of food purchases, our findings support the potential of spectrum labels to marginally enhance diet quality.

Explanation:

Linking the current study to existing research strengthens the argument for adopting spectrum labels.

Suggestion 10

Location: Introduction, Paragraph 258

Category: significance

Focus: impact

Original Text:

Together with prior research, our results suggest that if FDA pursues a high-in label, designs more similar to labels in Latin America may be more effective.

Improved Version:

Building on existing research, our findings indicate that for FDA to consider a high-in label, designs akin to those in Latin America could yield greater effectiveness.

Explanation:

Strengthening the connection to previous studies enhances the recommendation.

Suggestion 11

Location: Introduction, Paragraph 269

Category: significance

Focus: impact

Original Text:

Future research should explore how to design spectrum labels such that they are linked to federal regulations, such as federal guidelines for nutrient content or the FDA's definition of "healthy."

Improved Version:

Subsequent studies should investigate strategies to align spectrum labels with federal regulations, incorporating federal nutrient content guidelines or the FDA's 'healthy' definition.

Explanation:

Providing specific directions for future research enhances the clarity of the recommendation.

Detailed Feedback

Context Analysis

The introduction sets the stage by highlighting the impact of poor diet quality on health and the need for accessible nutrition information.

Problem Analysis

The introduction effectively identifies the lack of clarity on the effectiveness of different front-of-package labels in promoting healthier food choices.

Objectives Analysis

The objectives are clearly stated, focusing on evaluating the impact of various front-of-package labels on food purchases.

Significance Assessment

The introduction effectively justifies the research by emphasizing the potential of spectrum labels to improve diet quality and reduce health disparities.

Structure Evaluation

The introduction is well-organized, leading from background information to the research gap, objectives, and significance of the study.

S4 - Literature Review

Score: 7/10

Summary

The literature review provides a good foundation for the study but could benefit from improvements in coverage breadth, critical analysis, and integration with existing research. Enhancing the organization, updating citations, and improving the synthesis of findings would elevate the quality and impact of the literature review.

Critical Remarks

Remark 1

Category: coverage

Location: Introduction

Issue: Limited coverage of historical context regarding previous front-of-package labeling systems and their impact on consumer behavior.

Severity: medium

Impact: Reduces the comprehensiveness of the literature review and weakens the foundation for the current study.

Remark 2

Category: analysis

Location: Results

Issue: Lack of in-depth critical analysis on why certain labeling systems performed better than others.

Severity: medium

Impact: Weakens the interpretation of results and limits the insights derived from the study.

Remark 3

Category: structure

Location: Methods

Issue: The methodological review lacks clarity in explaining the rationale behind the selection of specific labeling systems for comparison.

Severity: low

Impact: A clearer explanation would enhance the transparency and reproducibility of the study.

Remark 4

Category: citations

Location: References

Issue: Limited recent citations on front-of-package labeling research, missing out on the latest developments in the field.

Severity: medium

Impact: Reduces the currency and relevance of the literature review.

Remark 5

Category: integration

Location: Discussion

Issue: Insufficient integration of the study findings with existing research on front-of-package labeling effectiveness.

Severity: high

Impact: Weakens the overall contribution of the study to the existing body of knowledge.

Improvement Suggestions

Suggestion 1

Location: Introduction

Category: coverage

Focus: breadth

Original Text:

Almost no studies have evaluated these Nutrition Info labels.

Improved Version:

Limited research has evaluated the effectiveness of Nutrition Info labels in influencing consumer behavior.

Explanation:

Provides a clearer statement on the existing research gap and the need for further investigation.

Suggestion 2

Location: Results

Category: analysis

Focus: synthesis

Original Text:

The effects of the labels on overall healthfulness were not moderated by nutrition literacy, income, or educational attainment.

Improved Version:

The study found no significant moderation effects of nutrition literacy, income, or educational attainment on the impact of labeling systems on overall healthfulness.

Explanation:

Clarifies the statistical findings and enhances the precision of the statement.

Suggestion 3

Location: Methods

Category: structure

Focus: organization

Original Text:

We randomized 6,002 participants; of these, 5,636 participants were included in the analyses.

Improved Version:

Out of the 6,002 participants randomized, 5,636 were included in the analyses.

Explanation:

Improves clarity and consistency in reporting participant numbers.

Suggestion 4

Location: Discussion

Category: coverage

Focus: relevance

Original Text:

The spectrum labels led to overall healthier selections compared to all other labeling systems we tested.

Improved Version:

Overall, the spectrum labels resulted in healthier selections compared to all other tested labeling systems.

Explanation:

Enhances readability and coherence in presenting study findings.

Suggestion 5

Location: Methods

Category: coverage

Focus: depth

Original Text:

The Nutrition Info labels we studied used traffic-light color coding to convey when foods were low (green), medium (yellow), or high (red) in nutrients of concern.

Improved Version:

The Nutrition Info labels in the study employed traffic-light color coding to indicate nutrient levels: green for low, yellow for medium, and red for high.

Explanation:

Provides a clearer description of the labeling system for better understanding.

Detailed Feedback

Coverage Analysis

The literature review provides a good overview of the current state of front-of-package labeling research but lacks depth in historical context and coverage of recent studies. Expanding the coverage breadth would enhance the foundation for the study.

Analysis Quality

The critical analysis of the study results is somewhat limited, with opportunities to delve deeper into the reasons behind the effectiveness of certain labeling systems. A more thorough analysis would strengthen the interpretation of findings.

Structure Evaluation

The organization of the literature review is logical, but some sections, such as the methodological review, could benefit from clearer explanations to improve transparency and reproducibility.

Citation Assessment

While the review includes relevant citations, there is a lack of recent references, impacting the currency and relevance of the study. Updating the citations with recent research would enhance the quality of the literature review.

Integration Review

The integration of the study findings with existing research is somewhat lacking, reducing the overall impact and contribution of the study to the field. Strengthening the connections with prior literature would improve the significance of the study.

S5 - Methodology

Score: 7/10

Summary

Overall, the study demonstrates a robust experimental design with a focus on evaluating the impact of front-of-package labeling systems on consumer behavior. While there are areas for improvement in providing clearer rationale for the chosen labels, detailing statistical methods, and enhancing the discussion of ethical considerations, the study's findings offer valuable insights into the effectiveness of different labeling approaches. Addressing these improvement suggestions would enhance the quality and rigor of the research.

Critical Remarks

Remark 1

Category: design

Location: Introduction

Issue: Lack of detailed rationale for selecting specific labeling systems to compare against FDA's proposed Nutrition Info labels.

Severity: medium

Impact: Weakens the justification for the chosen research design.

Remark 2

Category: analysis

Location: Methods

Issue: Limited explanation of the statistical methods used for analysis, potentially leading to ambiguity in result interpretation.

Severity: medium

Impact: May hinder the reproducibility and robustness of the study findings.

Improvement Suggestions

Suggestion 1

Location: Introduction

Category: design

Focus: rationale

Original Text:

Almost no studies have evaluated these Nutrition Info labels.

Improved Version:

Limited research has assessed the effectiveness of Nutrition Info labels in influencing consumer behavior.

Explanation:

Provides a more nuanced and accurate statement about the existing research gap.

Suggestion 2

Location: Methods

Category: methods

Focus: procedures

Original Text:

We assessed the effects of 6 front-of-package labeling systems...

Improved Version:

The study evaluated the impact of six distinct front-of-package labeling systems...

Explanation:

Enhances clarity and precision in describing the research scope.

Suggestion 3

Location: Analysis Methods

Category: analysis

Focus: statistical methods

Original Text:

We regressed healthfulness on indicator variables for each trial arm...

Improved Version:

Healthfulness was regressed against indicator variables representing each trial arm...

Explanation:

Improves the clarity and specificity of the statistical analysis description.

Suggestion 4

Location: Procedures

Category: methods

Focus: procedures

Original Text:

The primary outcome was the healthfulness of participants' selections...

Improved Version:

The primary outcome focused on the healthfulness of participants' food and beverage choices...

Explanation:

Clarifies the primary outcome measure for better understanding.

Suggestion 5

Location: Analysis Methods

Category: analysis

Focus: statistical methods

Original Text:

The effects of the labels on overall healthfulness were not moderated by nutrition literacy, income, or educational attainment.

Improved Version:

There was no significant moderation effect of nutrition literacy, income, or educational attainment on the impact of the labels on overall healthfulness.

Explanation:

Provides a more precise and detailed description of the moderation analysis results.

Suggestion 6

Location: Methods

Category: methods

Focus: procedures

Original Text:

We estimated sample size needs to detect a difference in healthfulness...

Improved Version:

Sample size requirements were estimated to detect differences in healthfulness...

Explanation:

Enhances clarity by specifying the purpose of sample size estimation.

Detailed Feedback

Design Analysis

The research design lacks a detailed rationale for selecting specific labeling systems for comparison, which weakens the justification for the study's focus. Providing a more robust explanation of why these specific labels were chosen would enhance the design's clarity and validity.

Methods Assessment

The methodology section would benefit from a more detailed explanation of the statistical methods used for analysis. Providing a clear description of the analytical techniques employed would improve the transparency and reproducibility of the study.

Analysis Evaluation

The analysis methods utilized in the study need to be more explicitly outlined to ensure the robustness and reliability of the results. Enhancing the clarity and specificity of the analytical approach would strengthen the interpretation of the findings.

Quality Review

The study demonstrates a strong focus on validity and reliability through the use of specific outcome measures and statistical analyses. However, further details on the validity measures and reliability assessments employed would enhance the methodological rigor of the research.

Ethics Compliance

The study appears to have addressed ethical considerations by obtaining institutional review board approval and ensuring participant consent. However, a more comprehensive discussion of the ethical implications of the research, particularly regarding participant privacy and data handling, would strengthen the ethical framework of the study.

S6 - Results

Score: 7/10

Summary

The study presents valuable insights into the impact of different front-of-package labeling systems on food and beverage selections. While the statistical analysis is robust and the study design is commendable, there are areas for improvement in data presentation, result interpretation, and clarity of reporting. Enhancing the presentation of participant

characteristics, providing deeper insights into the implications of the findings, and improving the clarity of figures and tables would strengthen the overall quality of the study.

Critical Remarks

Remark 1

Category: presentation

Location: Table 1

Issue: The table lacks clarity in presenting participant characteristics across different labeling systems.

Severity: medium

Impact: Affects the understanding of participant demographics and their distribution among different labeling groups.

Remark 2

Category: interpretation

Location: Results section

Issue: The interpretation of the results lacks depth and fails to provide a comprehensive analysis of the implications.

Severity: medium

Impact: Limits the understanding of the significance and broader implications of the study findings.

Improvement Suggestions

Suggestion 1

Location: Table 1

Category: presentation

Focus: clarity

Original Text:

The table lacks clarity in presenting participant characteristics across different labeling systems.

Improved Version:

Enhance Table 1 by providing a clearer layout with distinct columns for each labeling system to facilitate easy comparison of participant characteristics.

Explanation:

Improving the table layout will enhance readability and aid in understanding participant demographics.

Suggestion 2

Location: Results section

Category: interpretation

Focus: meaning

Original Text:

The interpretation of the results lacks depth and fails to provide a comprehensive analysis of the implications.

Improved Version:

Enhance the result interpretation by discussing the broader implications of the findings on public health policies and consumer behavior.

Explanation:

A more comprehensive analysis will provide a clearer understanding of the study's impact and relevance.

Suggestion 3

Location: Results section

Category: interpretation

Focus: implications

Original Text:

The effects of the labels on overall healthfulness were not moderated by nutrition literacy, income, or educational attainment.

Improved Version:

The effects of the labels on overall healthfulness were consistent across different levels of nutrition literacy, income, and educational attainment.

Explanation:

Clarifying the consistency of effects across different demographic variables enhances result interpretation.

Suggestion 4

Location: Results section

Category: interpretation

Focus: meaning

Original Text:

The spectrum labels led to overall healthier selections compared to all other labeling systems we tested.

Improved Version:

The spectrum labels consistently resulted in healthier selections compared to all other labeling systems tested in the study.

Explanation:

Adding consistency in reporting strengthens the interpretation of the spectrum labels' effectiveness.

Suggestion 5

Location: Results section

Category: interpretation

Focus: implications

Original Text:

The Nutrition Info labels did not improve healthfulness compared to the positive labels.

Improved Version:

The Nutrition Info labels did not show a significant improvement in healthfulness compared to the positive labels.

Explanation:

Adding clarity on the significance of the findings enhances the interpretation of the Nutrition Info labels' impact.

Suggestion 6

Location: Results section

Category: interpretation

Focus: meaning

Original Text:

The high-in labels did not lead to healthier selections than positive labels.

Improved Version:

The high-in labels did not demonstrate a significant improvement in healthfulness compared to positive labels.

Explanation:

Clarifying the lack of significant impact enhances the interpretation of the high-in labels' effectiveness.

Suggestion 7

Location: Results section

Category: interpretation

Focus: implications

Original Text:

The spectrum labels outperformed the positive labels for most secondary nutrition outcomes.

Improved Version:

The spectrum labels showed superior performance in most secondary nutrition outcomes compared to the positive labels.

Explanation:

Enhancing the comparison language strengthens the interpretation of the spectrum labels' effectiveness.

Suggestion 8

Location: Results section

Category: interpretation

Focus: meaning

Original Text:

The intervention labels generally outperformed the positive labels on psychological outcomes.

Improved Version:

The intervention labels generally exhibited better performance on psychological outcomes compared to the positive labels.

Explanation:

Improving the comparison language enhances the interpretation of the intervention labels' impact on psychological outcomes.

Suggestion 9

Location: Results section

Category: interpretation

Focus: implications

Original Text:

The effects of the food labels on healthfulness did not vary by nutrition literacy, income, or educational attainment.

Improved Version:

The impact of the food labels on healthfulness remained consistent across different levels of nutrition literacy, income, and educational attainment.

Explanation:

Clarifying the consistency of effects across demographic variables strengthens the interpretation of the study results.

Suggestion 10

Location: Results section

Category: interpretation

Focus: meaning

Original Text:

The spectrum labels may be more promising than both existing positive labels and the FDA's proposed Nutrition Info labels for promoting healthier purchases.

Improved Version:

The spectrum labels show promise in promoting healthier purchases compared to both existing positive labels and the FDA's proposed Nutrition Info labels.

Explanation:

Enhancing the comparison language clarifies the potential of the spectrum labels in improving food choices.

Detailed Feedback

Presentation Analysis

The data presentation in Table 1 could be improved by providing a clearer layout for participant characteristics. The statistical analysis is robust, utilizing regression models to assess the effects of different labeling systems on healthfulness.

Analysis Quality

The statistical analysis methods employed, such as OLS regression and interaction testing, are appropriate for evaluating the impact of front-of-package labeling systems. The study's quality is enhanced by the use of a large and diverse sample in a realistic online grocery store setting.

Interpretation Review

The interpretation of results could be strengthened by providing deeper insights into the implications of the findings on public health policies and consumer behavior. A more comprehensive discussion of the broader implications would enhance the overall interpretation of the study results.

Visualization Assessment

The figures and tables, such as Figure 1 and Table 1, could benefit from improved clarity and organization to facilitate easier understanding of participant characteristics and labeling system comparisons.

Significance Evaluation

The reporting of statistical significance, effect sizes, and confidence intervals is generally clear. However, a more detailed discussion on the significance of the results and their practical implications would enhance the reporting of statistical findings.

S7 - Discussion

Score: 8/10

Summary

The discussion provides a clear interpretation of the study results on front-of-package labeling systems' impact on food choices. However, there are notable gaps in literature comparison, bias acknowledgment, and specificity in future research recommendations. Enhancing the discussion with a more comprehensive literature review, addressing biases, providing specific future research directions, and deepening the practical and theoretical implications would significantly improve the overall quality and coherence of the study.

Critical Remarks

Remark 1

Category: context

Location: Introduction

Issue: The discussion lacks a comprehensive comparison with existing literature on front-of-package labeling systems. There is a need to provide more detailed insights into how the current study aligns with or contradicts previous research.

Severity: medium

Impact: This omission weakens the contextualization of the study within the broader research landscape.

Remark 2

Category: reflection

Location: Limitations

Issue: The limitations section does not address potential biases in the study design or data collection methods. It is essential to acknowledge and discuss any biases that may have influenced the results.

Severity: high

Impact: Failure to address biases undermines the credibility and validity of the study findings.

Remark 3

Category: reflection

Location: Limitations

Issue: The discussion on future work is limited to general statements. It would be beneficial to provide specific recommendations for future research directions based on the study findings.

Severity: medium

Impact: Enhancing the future work section with specific suggestions can guide researchers in building on the current study.

Improvement Suggestions

Suggestion 1

Location: Results - Discussion of moderation

Category: interpretation

Focus: interpretation

Original Text:

The effects of the labels on overall healthfulness were not moderated by nutrition literacy, income, or educational attainment.

Improved Version:

The effects of the labels on overall healthfulness were consistent across different levels of nutrition literacy, income, and educational attainment.

Explanation:

Clarity in stating the consistency of effects across various demographic factors.

Suggestion 2

Location: Results - Conclusion

Category: interpretation

Focus: interpretation

Original Text:

The Nutrition Info labels did not improve healthfulness compared to the positive labels.

Improved Version:

Contrary to expectations, the Nutrition Info labels did not lead to significant improvements in healthfulness compared to the positive labels.

Explanation:

Adding context to the unexpected findings for better result interpretation.

Suggestion 3

Location: Introduction

Category: context

Focus: comparison

Original Text:

The study aimed to evaluate whether the Nutrition Info label and other new front-of-package labels lead to healthier purchases compared to positive labels and compared to one another.

Improved Version:

The study aimed to assess the impact of the Nutrition Info label and other new front-of-package labels on the healthfulness of purchases, both in comparison to positive labels and among themselves.

Explanation:

Clarifying the study's specific focus and objectives for better coherence.

Suggestion 4

Location: Limitations

Category: reflection

Focus: limitations

Original Text:

The limitations section is brief and lacks a comprehensive discussion of potential biases in the study design.

Improved Version:

The limitations section should be expanded to include a thorough examination of potential biases inherent in the study design, such as selection bias or measurement errors.

Explanation:

Enhancing the limitations discussion to acknowledge and address potential sources of bias.

Suggestion 5

Location: Practical Implications

Category: impact

Focus: implications

Original Text:

The discussion on practical implications is limited to general statements.

Improved Version:

To enhance the practical implications, specific recommendations for policymakers and food industry stakeholders should be provided based on the study findings.

Explanation:

Adding actionable insights for stakeholders to apply the research findings.

Suggestion 6

Location: Theoretical Contributions

Category: impact

Focus: implications

Original Text:

Theoretical contributions are briefly mentioned without a detailed exploration of how the study findings contribute to existing theory.

Improved Version:

To strengthen theoretical contributions, a deeper analysis of how the study results align with or challenge current theoretical frameworks in nutrition labeling research is needed.

Explanation:

Deepening the discussion on theoretical implications for advancing the field of nutrition labeling.

Detailed Feedback**Interpretation Analysis**

The interpretation of results is clear and concise, highlighting the key findings related to the effectiveness of different front-of-package labeling systems on promoting healthier food choices.

Context Review

The literature comparison is limited, with minimal references to previous studies on front-of-package labeling systems. A more comprehensive comparison with existing research could provide a stronger foundation for the study's significance.

Reflection Assessment

The limitations section lacks a thorough discussion of potential biases and methodological limitations. Additionally, the future work section could benefit from more specific recommendations for future research directions based on the study outcomes.

Impact Evaluation

The practical implications and theoretical contributions are briefly discussed, but there is room for deeper exploration of how the study findings can inform policy decisions and contribute to theoretical advancements in the field of nutrition labeling.

Quality Analysis

The discussion is well-structured and logically presented, but it would benefit from more in-depth analysis and contextualization within the broader literature. Addressing biases, providing specific future research directions, and enhancing the practical and theoretical implications would strengthen the overall quality of the discussion.

S8 - Conclusion

Score: 7/10

Summary

The conclusion provides valuable insights into the effectiveness of different front-of-package labeling systems on promoting healthier food choices. However, there are areas

that require improvement, such as providing more detailed analysis on the results, connecting the findings back to the research objectives, exploring implications more thoroughly, enhancing presentation clarity, and explicitly stating the study's contribution to the field.

Critical Remarks

Remark 1

Category: support

Location: Results section

Issue: Lack of detailed analysis on why certain labels did not lead to healthier purchases compared to positive labels

Severity: medium

Impact: Weakens the credibility of the conclusions drawn

Remark 2

Category: objectives

Location: Introduction and Conclusion

Issue: Limited discussion on how the study fulfills the research objectives

Severity: medium

Impact: Reduces clarity on the study's purpose and outcomes

Remark 3

Category: implications

Location: Implications section

Issue: Insufficient exploration of practical and theoretical implications of the findings

Severity: medium

Impact: Diminishes the significance of the study's contributions

Remark 4

Category: presentation

Location: Throughout the text

Issue: Lack of concise and clear language in some sections

Severity: low

Impact: Affects readability and understanding

Improvement Suggestions

Suggestion 1

Location: Results section

Category: presentation

Focus: clarity

Original Text:

Neither the Nutrition Info, positive + Nutrition Info, high-in, nor positive + high-in labels led participants to make healthier purchases compared to the positive labels only.

Improved Version:

Participants did not make healthier purchases when exposed to Nutrition Info, positive + Nutrition Info, high-in, or positive + high-in labels compared to positive labels only.

Explanation:

Clarifies the comparison and improves readability.

Suggestion 2

Location: Results section

Category: presentation

Focus: clarity

Original Text:

The spectrum labels led participants to make healthier purchases compared to both the positive labels only and all other labeling systems.

Improved Version:

Participants made healthier purchases with spectrum labels compared to both positive labels only and all other labeling systems.

Explanation:

Enhances clarity and removes redundancy.

Suggestion 3

Location: Results section

Category: presentation

Focus: clarity

Original Text:

The effects of the labels on healthfulness did not vary by nutrition literacy, income, or educational attainment.

Improved Version:

The effects of the labels on healthfulness were consistent across different levels of nutrition literacy, income, and educational attainment.

Explanation:

Provides a clearer statement and improves precision.

Suggestion 4

Location: Discussion section

Category: presentation

Focus: clarity

Original Text:

In this study, Nutrition Info labels similar to FDA's proposed labels did not lead to healthier selections than positive labels.

Improved Version:

Nutrition Info labels, similar to FDA's proposed labels, did not result in healthier selections compared to positive labels.

Explanation:

Enhances sentence structure and readability.

Suggestion 5

Location: Discussion section

Category: presentation

Focus: clarity

Original Text:

In this study, the Nutrition Info, high-in, and spectrum labels all increased the likelihood that participants correctly identified healthier foods compared to the status quo of positive labels.

Improved Version:

The Nutrition Info, high-in, and spectrum labels increased the likelihood that participants correctly identified healthier foods compared to the positive labels.

Explanation:

Simplifies the sentence structure for better comprehension.

Suggestion 6

Location: Discussion section

Category: presentation

Focus: clarity

Original Text:

The spectrum labels led to overall healthier selections compared to all other labeling systems we tested.

Improved Version:

Spectrum labels resulted in overall healthier selections compared to all other tested labeling systems.

Explanation:

Improves sentence flow and readability.

Suggestion 7

Location: Discussion section

Category: presentation

Focus: clarity

Original Text:

The Nutrition Info labels we studied used traffic-light color coding to convey when foods were low (green), medium (yellow), or high (red) in nutrients of concern.

Improved Version:

The Nutrition Info labels in our study utilized traffic-light color coding to indicate low (green), medium (yellow), or high (red) levels of nutrients of concern.

Explanation:

Enhances specificity and clarity.

Suggestion 8

Location: Discussion section

Category: support

Focus: evidence

Original Text:

Our results align with prior studies finding that spectrum labels improve the healthfulness of food and beverage purchases.

Improved Version:

Our findings are consistent with previous research indicating that spectrum labels enhance the healthfulness of food and beverage choices.

Explanation:

Strengthens the connection to existing literature.

Suggestion 9

Location: Discussion section

Category: implications

Focus: future_directions

Original Text:

Together with prior research, our results suggest that if FDA pursues a high-in label, designs more similar to labels in Latin America may be more effective.

Improved Version:

In conjunction with previous studies, our findings imply that FDA should consider designs akin to Latin American labels for more effective high-in labeling.

Explanation:

Enhances the recommendation based on existing evidence.

Suggestion 10

Location: Discussion section

Category: objectives

Focus: future_directions

Original Text:

Future research should explore how to design spectrum labels such that they are linked to federal regulations, such as federal guidelines for nutrient content or the FDA's definition of 'healthy.'

Improved Version:

Subsequent studies should investigate strategies to align spectrum labels with federal regulations, such as nutrient content guidelines or the FDA's 'healthy' definition.

Explanation:

Provides a clear direction for future research.

Detailed Feedback

Support Analysis

The results section provides a clear comparison of different labeling systems but lacks in-depth analysis on why certain labels did not lead to healthier purchases. More detailed explanations could strengthen the support for the conclusions.

Objective Fulfillment

While the study's objectives are outlined in the abstract and introduction, there is limited discussion on how the study fulfills these objectives in the conclusion. Connecting the findings back to the research objectives would enhance clarity.

Implications Analysis

The implications section briefly touches on practical and theoretical implications but lacks depth. Further exploration of the practical and theoretical implications of the findings would strengthen the significance of the study's contributions.

Presentation Analysis

The text could benefit from more concise and clear language in some sections to enhance readability and understanding. Improving the presentation clarity would make the conclusions more impactful.

Contribution Analysis

The study's contribution to the field is evident, but the clarity of this contribution could be enhanced by explicitly stating how the findings advance current knowledge and practice.

S9 - References

Score: 7/10

Summary

The references in the manuscript show some issues related to accuracy, completeness, format, and quality. Improvements are needed in providing complete reference details, ensuring citation-text alignment, and enhancing the relevance and quality of the sources. Additionally, restructuring sentences for clarity and consistency in formatting will enhance the overall quality of the references.

Critical Remarks

Remark 1

Category: accuracy

Location: Reference list organization

Issue: Incomplete reference details for some sources, missing publication dates

Severity: medium

Impact: Affects the credibility and replicability of the research

Remark 2

Category: completeness

Location: Citation accuracy

Issue: Inconsistencies in author affiliations and missing DOI numbers

Severity: medium

Impact: May hinder readers from accessing the original sources

Remark 3

Category: format

Location: Citation-text alignment

Issue: Lack of direct citations in some sections of the text

Severity: low

Impact: Weakens the connection between text and supporting references

Remark 4

Category: quality

Location: Source relevance

Issue: Limited discussion on the relevance of some references to the study

Severity: medium

Impact: Weakens the foundation of the research

Improvement Suggestions

Suggestion 1

Location: Reference list organization

Category: completeness

Focus: citation

Original Text:

Grummon AH, Musicus AA, Moran AJ, Salvia MG, Rimm EB.

Improved Version:

Grummon AH, Musicus AA, Moran AJ, Salvia MG, Rimm EB. (2023).

Explanation:

Adding publication year enhances reference completeness and allows for easier tracking of sources.

Suggestion 2

Location: Citation accuracy

Category: completeness

Focus: citation

Original Text:

United States Food and Drug Administration Human Foods Program.

Improved Version:

United States Food and Drug Administration Human Foods Program. (2024).

Explanation:

Including the publication year provides essential information for source identification.

Suggestion 3

Location: Citation-text alignment

Category: completeness

Focus: reference

Original Text:

eFigure 1 shows an example page.

Improved Version:

Figure 1 illustrates an example page (Grummon et al., 2023).

Explanation:

Linking the figure to the reference enhances citation-text alignment.

Suggestion 4

Location: Introduction

Category: accuracy

Focus: citation

Original Text:

Almost no studies have evaluated these Nutrition Info labels.

Improved Version:

Limited research has assessed the effectiveness of Nutrition Info labels (Grummon et al., 2023).

Explanation:

Providing a direct citation strengthens the statement and aligns it with supporting evidence.

Suggestion 5

Location: Discussion

Category: accuracy

Focus: citation

Original Text:

The spectrum labels led to overall healthier selections compared to all other labeling systems we tested.

Improved Version:

The study findings indicated that spectrum labels resulted in healthier selections compared to all other tested labeling systems (Grummon et al., 2023).

Explanation:

Including the study reference enhances the credibility of the statement.

Suggestion 6

Location: Methods

Category: format

Focus: style

Original Text:

First, we examined the effects of the front-of-package labels on the primary outcome of healthfulness...

Improved Version:

Initially, the study assessed the impact of front-of-package labels on the primary outcome of healthfulness...

Explanation:

Restructuring the sentence for clarity and conciseness.

Suggestion 7

Location: Results

Category: format

Focus: style

Original Text:

We randomized 6,002 participants; of these, 5,636 participants were included in the analyses...

Improved Version:

Out of 6,002 participants randomized, 5,636 were included in the analyses...

Explanation:

Rephrasing for improved readability and flow.

Suggestion 8

Location: Introduction

Category: quality

Focus: style

Original Text:

In this randomized trial in a large sample of US adults...

Improved Version:

The randomized trial conducted in a large sample of US adults...

Explanation:

Enhancing the introductory statement for clarity and precision.

Suggestion 9

Location: Results

Category: accuracy

Focus: style

Original Text:

The effects of the labels on healthfulness did not vary by nutrition literacy, income, or educational attainment.

Improved Version:

There was no significant variation in the effects of the labels on healthfulness based on nutrition literacy, income, or educational attainment.

Explanation:

Providing a more detailed and precise statement.

Suggestion 10

Location: Discussion

Category: quality

Focus: style

Original Text:

Together, these results suggest that adopting spectrum labels could improve diet quality without exacerbating health disparities.

Improved Version:

The collective findings indicate that implementing spectrum labels may enhance diet quality without worsening health disparities.

Explanation:

Enhancing the conclusion for clarity and impact.

Detailed Feedback**Accuracy Analysis**

The references lack complete details such as publication years and DOI numbers, affecting the accuracy of citations.

Completeness Analysis

Some references have inconsistencies in author affiliations and missing DOI numbers, impacting the completeness of the sources.

Format Analysis

There are instances where direct citations are missing in the text, leading to format inconsistencies.

Quality Analysis

Limited discussion on the relevance of some references weakens the quality of the sources used in the study.

Organization Analysis

The reference list lacks consistent formatting and complete details, affecting the overall organization of the references.

S10 - Supplementary Materials

Score: 7/10

Summary

The supplementary materials offer a comprehensive and detailed analysis of the study, providing relevant information on the research design, methods, results, and implications. While there are areas for improvement in terms of clarity and conciseness in certain sections, overall, the supplementary materials enhance the understanding and completeness of the study findings.

Critical Remarks

Remark 1

Category: relevance

Location: ABSTRACT

Issue: The abstract lacks a clear statement on the implications of the study findings for policy or practice.

Severity: medium

Impact: This omission reduces the overall impact of the study and may hinder the reader's understanding of the significance of the results.

Remark 2

Category: clarity

Location: METHODS

Issue: The description of the setting and procedures is overly detailed, making it challenging for readers to grasp the key methodological aspects quickly.

Severity: medium

Impact: This may lead to reader confusion and detract from the overall clarity of the methodology section.

Improvement Suggestions

Suggestion 1

Location: RESULTS

Category: clarity

Focus: presentation

Original Text:

The effects of the labels on overall healthfulness were not moderated by nutrition literacy, income, or educational attainment.

Improved Version:

The effects of the labels on overall healthfulness were consistent across different levels of nutrition literacy, income, and educational attainment.

Explanation:

Enhances clarity by providing a more concise and direct statement.

Suggestion 2

Location: METHODS

Category: clarity

Focus: presentation

Original Text:

We assessed healthfulness using the United Kingdom's Ofcom Nutrient Profiling Model scores.

Improved Version:

Healthfulness was assessed using the United Kingdom's Ofcom Nutrient Profiling Model scores.

Explanation:

Improves sentence structure and readability by removing unnecessary words.

Suggestion 3

Location: METHODS

Category: clarity

Focus: presentation

Original Text:

We calculated average Ofcom scores across all participants' selections in the shopping task, weighted by the number of servings in each product selected.

Improved Version:

Average Ofcom scores were calculated across all participants' selections in the shopping task, weighted by the number of servings in each product selected.

Explanation:

Enhances clarity by restructuring the sentence for better flow and readability.

Suggestion 4

Location: METHODS

Category: clarity

Focus: presentation

Original Text:

The primary outcome was the healthfulness of participants' selections in the shopping task.

Improved Version:

The primary outcome focused on the healthfulness of participants' selections in the shopping task.

Explanation:

Adds specificity and clarity to the description of the primary outcome.

Suggestion 5

Location: METHODS

Category: clarity

Focus: presentation

Original Text:

The trial took place in a naturalistic online grocery store.

Improved Version:

The trial was conducted in a naturalistic online grocery store.

Explanation:

Improves sentence structure and clarity by using a more active voice.

Suggestion 6

Location: METHODS

Category: clarity

Focus: presentation

Original Text:

The Nutrition Info labels we studied used traffic-light color coding to convey when foods were low (green), medium (yellow), or high (red) in nutrients of concern.

Improved Version:

The Nutrition Info labels used traffic-light color coding to indicate whether foods were low (green), medium (yellow), or high (red) in nutrients of concern.

Explanation:

Clarifies the purpose of the color coding for better understanding.

Suggestion 7

Location: RESULTS

Category: clarity

Focus: presentation

Original Text:

The effects of the labels on healthfulness were not moderated by nutrition literacy, income, or educational attainment.

Improved Version:

The effects of the labels on healthfulness were consistent across different levels of nutrition literacy, income, and educational attainment.

Explanation:

Enhances clarity by providing a more concise and direct statement.

Suggestion 8

Location: METHODS

Category: clarity

Focus: presentation

Original Text:

The Nutrition Info labels we studied used traffic-light color coding to convey when foods were low (green), medium (yellow), or high (red) in nutrients of concern.

Improved Version:

The Nutrition Info labels used traffic-light color coding to indicate whether foods were low (green), medium (yellow), or high (red) in nutrients of concern.

Explanation:

Clarifies the purpose of the color coding for better understanding.

Suggestion 9

Location: METHODS

Category: clarity

Focus: presentation

Original Text:

The primary outcome was the healthfulness of participants' selections in the shopping task.

Improved Version:

The primary outcome focused on the healthfulness of participants' selections in the shopping task.

Explanation:

Adds specificity and clarity to the description of the primary outcome.

Suggestion 10

Location: METHODS

Category: clarity

Focus: presentation

Original Text:

The trial took place in a naturalistic online grocery store.

Improved Version:

The trial was conducted in a naturalistic online grocery store.

Explanation:

Improves sentence structure and clarity by using a more active voice.

Detailed Feedback**Relevance Analysis**

The supplementary materials provide detailed information on the study design, methods, results, and implications, offering a comprehensive overview of the research conducted. The study's relevance to the main text is evident through the alignment of objectives and outcomes.

Clarity Analysis

The presentation of data and methodological details is clear and concise, aiding in the understanding of the study procedures and outcomes. The use of tables and figures enhances the clarity of data presentation.

Consistency Analysis

The supplementary materials maintain consistency with the main text by elaborating on the study's objectives, methods, and results in a structured manner. The alignment between the main text and supplementary materials is evident throughout the document.

Completeness Analysis

The supplementary materials provide a thorough description of the study methodology, data analysis, and results, ensuring the completeness of information presented. Methodological details are sufficiently detailed to allow for replication and understanding of the study.

Rigor Agents (R1-R7)**R1 - Originality and Contribution**

Score: 8/10

Summary

Overall, the study presents a novel research approach in evaluating front-of-package labeling systems and makes unique contributions by highlighting the effectiveness of spectrum labels in promoting healthier food choices. However, improvements are needed in clearly articulating the novelty, discussing the specific contributions of each labeling

system, verifying the novelty claims, providing a more detailed comparison with existing literature, and enhancing the clarity of the field advancement implications.

Critical Remarks

Remark 1

Category: novelty

Location: Introduction

Issue: Lack of clear identification of the novel research approach or methodology.

Severity: medium

Impact: Weakens the overall originality of the study.

Remark 2

Category: contribution

Location: Results

Issue: Limited discussion on the unique contributions of each front-of-package labeling system.

Severity: medium

Impact: Reduces clarity on the specific impacts of different labels.

Remark 3

Category: verification

Location: Discussion

Issue: Insufficient verification of the stated novelty claims regarding the spectrum labels.

Severity: medium

Impact: Raises questions about the validity of the conclusions.

Remark 4

Category: comparison

Location: Literature Review

Issue: Limited comparison with existing literature on the effectiveness of front-of-package labels.

Severity: medium

Impact: Lacks depth in contextualizing the study within the broader research landscape.

Remark 5

Category: advancement

Location: Conclusion

Issue: The conclusion lacks a comprehensive summary of the field advancement.

Severity: low

Impact: Reduces the clarity of the study's overall impact on the field.

Improvement Suggestions

Suggestion 1

Location: Introduction

Category: literature

Focus: comparison

Original Text:

Almost no studies have evaluated these Nutrition Info labels.

Improved Version:

Limited research has been conducted on the effectiveness of Nutrition Info labels.

Explanation:

Provides a more accurate representation of the existing literature gap.

Suggestion 2

Location: Methodology

Category: methodology

Focus: novelty

Original Text:

We assessed the effects of 6 front-of-package labeling systems...

Improved Version:

We comprehensively evaluated the impacts of six distinct front-of-package labeling systems...

Explanation:

Enhances the clarity and depth of the research approach description.

Suggestion 3

Location: Results

Category: results

Focus: contribution

Original Text:

The spectrum labels led participants to make healthier purchases compared to both the positive labels only...

Improved Version:

Spectrum labels significantly outperformed positive labels in promoting healthier food choices...

Explanation:

Provides a more impactful presentation of the contribution of spectrum labels.

Suggestion 4

Location: Conclusion

Category: conclusion

Focus: advancement

Original Text:

Together, these results suggest that adopting spectrum labels could improve diet quality without exacerbating health disparities.

Improved Version:

The findings indicate that implementing spectrum labels may enhance dietary quality while addressing health disparities.

Explanation:

Enhances the clarity and impact of the conclusion statement.

Detailed Feedback

Novelty Assessment

The study introduces a novel approach by evaluating the effectiveness of various front-of-package labeling systems on promoting healthier food choices. This research fills a gap in understanding the impact of different labeling strategies on consumer behavior.

Contribution Analysis

The study contributes valuable insights by demonstrating that spectrum labels are more effective in encouraging healthier food selections compared to other labeling systems. The results highlight the potential of spectrum labels to improve diet quality.

Verification Status

While the study claims novelty in evaluating Nutrition Info labels and spectrum labels, the verification of these claims could be strengthened by providing more in-depth analysis and comparison with existing literature.

Comparative Analysis

The literature review provides a basic comparison with existing work on front-of-package labeling but lacks a detailed analysis of the effectiveness of different labeling systems. A more comprehensive review of relevant studies could enhance the contextualization of the research.

Advancement Evaluation

The study advances knowledge by demonstrating the potential of spectrum labels to promote healthier food choices. However, a more robust discussion on the implications of

these findings for policy and public health initiatives could further enhance the advancement of knowledge in the field.

R2 - Impact and Significance

Score: 8/10

Summary

Overall, the study provides valuable insights into the impact of front-of-package labeling systems on consumer food choices. By highlighting the superior effectiveness of spectrum labels and identifying the limitations of existing and proposed systems, the research contributes significantly to the field of nutrition labeling and consumer behavior. However, there are areas for improvement in clearly articulating the study's influence on the field, discussing broader implications, and addressing future research directions.

Critical Remarks

Remark 1

Category: field_influence

Location: Introduction

Issue: The text lacks a clear statement on the potential influence of the study on the field of nutrition labeling and consumer behavior.

Severity: medium

Impact: This omission reduces the clarity of the study's significance and may hinder its recognition in the academic community.

Remark 2

Category: implications

Location: Discussion

Issue: The broader implications of the findings are not extensively discussed, limiting the understanding of the study's impact beyond the immediate results.

Severity: medium

Impact: This hinders the ability to draw comprehensive conclusions and may limit the study's relevance in informing future research and policy decisions.

Improvement Suggestions

Suggestion 1

Location: Introduction

Category: introduction

Focus: field_influence

Original Text:

The US requires most packaged foods to display the Nutrition Facts Label, but this label is displayed on the back or side of product packaging and includes only numerical information that most consumers have difficulty understanding.

Improved Version:

While the US mandates the Nutrition Facts Label on most packaged foods, its placement on the back or side of products and reliance on numerical data pose challenges for consumer comprehension.

Explanation:

Clarifies the issue and improves readability by breaking down complex information.

Suggestion 2

Location: Introduction

Category: introduction

Focus: field_influence

Original Text:

To address these gaps and inform policymaking, this study aimed to evaluate whether the Nutrition Info label and other new front-of-package labels lead to healthier purchases compared to positive labels and compared to one another.

Improved Version:

This study aimed to fill existing gaps and provide insights for policymakers by assessing the impact of Nutrition Info labels and other front-of-package labeling systems on promoting healthier food choices.

Explanation:

Streamlines the purpose statement and emphasizes the study's contribution to informing policy decisions.

Suggestion 3

Location: Results

Category: results

Focus: field_influence

Original Text:

The effects of the labels on healthfulness did not vary by nutrition literacy, income, or educational attainment.

Improved Version:

The study found consistent effects of the labels on healthfulness across different levels of nutrition literacy, income, and educational attainment.

Explanation:

Enhances clarity and highlights the robustness of the study findings.

Suggestion 4

Location: Discussion

Category: discussion

Focus: implications

Original Text:

In this study, Nutrition Info labels similar to FDA's proposed labels did not lead to healthier selections than positive labels, regardless of whether they were presented alone or alongside positive labels.

Improved Version:

Contrary to expectations, Nutrition Info labels, akin to FDA's proposed labels, did not result in healthier choices compared to positive labels, whether presented independently or in combination.

Explanation:

Provides a more concise and impactful summary of the findings.

Suggestion 5

Location: Discussion

Category: discussion

Focus: implications

Original Text:

The spectrum labels led to overall healthier selections compared to all other labeling systems we tested.

Improved Version:

Notably, spectrum labels consistently led to healthier choices compared to all other labeling systems examined in the study.

Explanation:

Emphasizes the significance of spectrum labels in promoting healthier food selections.

Suggestion 6

Location: Discussion

Category: discussion

Focus: future_research

Original Text:

Future research should explore how to design spectrum labels such that they are linked to federal regulations, such as federal guidelines for nutrient content or the FDA's definition of "healthy."

Improved Version:

Subsequent research endeavors could investigate strategies to align spectrum labels with federal regulations, such as incorporating federal nutrient guidelines or the FDA's definition of 'healthy'.

Explanation:

Provides a clear direction for future research and regulatory alignment.

Suggestion 7

Location: Results

Category: results

Focus: applications

Original Text:

The intervention labels generally outperformed the positive labels on psychological outcomes.

Improved Version:

Across psychological outcomes, the intervention labels consistently demonstrated superior performance compared to positive labels.

Explanation:

Highlights the impact of intervention labels on psychological factors.

Suggestion 8

Location: Discussion

Category: discussion

Focus: policy

Original Text:

FDA has emphasized that a core objective of mandating front-of-package food labels is to help consumers identify healthier foods.

Improved Version:

A key objective of FDA's front-of-package food labeling mandate is to assist consumers in recognizing healthier food choices.

Explanation:

Clarifies the regulatory goal and its implications for consumer decision-making.

Suggestion 9

Location: Conclusion

Category: conclusion

Focus: field_influence

Original Text:

Strengths of this study include the large and diverse sample, the randomized design, and the use of a realistic online grocery store.

Improved Version:

The study's strengths encompass a sizable and diverse sample, a randomized design, and the utilization of a realistic online grocery store.

Explanation:

Enhances the description of study strengths for better comprehension.

Suggestion 10

Location: Conclusion

Category: conclusion

Focus: future_research

Original Text:

Limitations include that participants had a single exposure to the labels in the context of an online store.

Improved Version:

One limitation is that participants were exposed to the labels only once within an online store setting.

Explanation:

Clarifies the study's limitations for better contextual understanding.

Detailed Feedback

Field Influence

The study provides valuable insights into the effectiveness of different front-of-package labeling systems on promoting healthier food choices. By demonstrating the superiority of spectrum labels over existing and proposed systems, the research contributes significantly to the field of nutrition labeling and consumer behavior.

Broader Implications

The findings suggest that adopting spectrum labels could have substantial implications for improving diet quality and consumer decision-making. Spectrum labels may offer a more effective approach to guiding consumers towards healthier food selections compared to current labeling practices.

Future Research Impact

The study highlights the need for further research to refine front-of-package labeling designs, align them with federal regulations, and explore the long-term effects of repeated

label exposure. Future studies could delve into the mechanisms underlying label effectiveness and investigate strategies to enhance consumer understanding and decision-making.

Practical Applications

The practical applications of this research extend to informing policymakers, food manufacturers, and retailers about the potential benefits of adopting spectrum labels. These labels could play a crucial role in promoting healthier food choices and improving public health outcomes. Additionally, the study underscores the importance of considering consumer behavior and preferences in designing effective front-of-package labeling systems.

Policy Implications

From a policy perspective, the study underscores the significance of regulatory interventions in shaping consumer food choices. The findings suggest that policymakers should consider the effectiveness of different labeling systems in influencing consumer behavior and prioritize evidence-based approaches to improve public health outcomes.

R3 - Ethics and Compliance

Score: 7/10

Summary

The study demonstrates a solid foundation in research ethics and compliance with ethical guidelines. However, improvements in data privacy measures, explicit consent procedures, and addressing potential biases are essential to enhance ethical standards and research integrity.

Critical Remarks

Remark 1

Category: privacy

Location: Methodology section

Issue: Lack of detailed information on data privacy measures for online grocery store participants.

Severity: medium

Impact: Poses a risk to data protection and confidentiality of participants' information.

Remark 2

Category: consent

Location: Methodology section

Issue: Limited information on the explicit consent process for data collection and analysis.

Severity: medium

Impact: Raises concerns about participants' understanding and voluntary participation.

Remark 3

Category: integrity

Location: Results section

Issue: Potential bias in the interpretation of results favoring spectrum labels without addressing limitations.

Severity: medium

Impact: Could undermine the credibility and objectivity of the study findings.

Improvement Suggestions

Suggestion 1

Location: Methodology section

Category: consent

Focus: consent

Original Text:

Participants were informed that 1 in 50 would be randomly chosen to receive their selections delivered to their home and the remainder of their budget as a gift card.

Improved Version:

Participants were clearly informed about the random selection process for receiving grocery deliveries and gift cards.

Explanation:

Enhances transparency and clarity in the consent process.

Suggestion 2

Location: Methodology section

Category: methodology

Focus: guidelines

Original Text:

Online stores like this one are realistic, acceptable, and valid settings for nutrition research studies.

Improved Version:

Online stores provide a realistic and valid setting for conducting nutrition research studies.

Explanation:

Clarifies the assertion and maintains accuracy in the statement.

Suggestion 3

Location: Results section

Category: results

Focus: integrity

Original Text:

The effects of the labels on healthfulness did not vary by nutrition literacy, income, or educational attainment.

Improved Version:

The effects of the labels on healthfulness were consistent across different levels of nutrition literacy, income, and educational attainment.

Explanation:

Enhances clarity and precision in reporting the findings.

Detailed Feedback

Conflicts Assessment

The authors have disclosed their affiliations and potential conflicts of interest adequately. However, further clarification on funding sources and sponsorships would enhance transparency.

Privacy Compliance

The study lacks explicit details on data privacy measures, including data encryption, storage, and access controls. Providing a comprehensive data privacy statement is essential to ensure participant confidentiality.

Consent Procedures

While the study mentions obtaining electronic informed consent, the process details, including information provided to participants and their rights, need to be explicitly outlined. Clear consent procedures are crucial for ethical research conduct.

Research Integrity

The study design and methodology appear robust, with a randomized trial approach. However, addressing potential biases and limitations in the interpretation of results would strengthen research integrity.

Guidelines Adherence

The study aligns with ethical guidelines in terms of participant recruitment and data collection. However, enhancing transparency in reporting results and addressing potential biases would further improve adherence to ethical standards.

R4 - Data and Code Availability

Score: 4/10

Summary

The study lacks clear statements on data availability and code repository, leading to challenges in reproducibility. Insufficient documentation and lack of access restriction justifications further hinder transparency. Detailed data presentation and code results are essential for reproducibility and should be improved to enhance the study's credibility.

Critical Remarks

Remark 1

Category: data_sharing

Location: Abstract

Issue: The abstract lacks a clear statement on data availability, making it difficult for readers to know if the data is accessible.

Severity: medium

Impact: This affects research transparency as it hinders the ability of other researchers to verify the findings.

Remark 2

Category: code_availability

Location: Methodology

Issue: There is no mention of a code repository for the implementation of the randomized trial, making it challenging for others to reproduce the study.

Severity: high

Impact: This significantly hampers the reproducibility of the study and limits the ability of other researchers to build upon the work.

Remark 3

Category: documentation

Location: Discussion

Issue: The discussion lacks detailed documentation on the methodology used for data

collection and code implementation, reducing the clarity of the research process.

Severity: medium

Impact: Insufficient documentation hinders the understanding of the study's methodology and limits the reproducibility of the results.

Remark 4

Category: restrictions

Location: Conclusion

Issue: There is no justification provided for any access restrictions to the data or code, leaving readers unaware of any limitations.

Severity: low

Impact: While the impact is low, providing access restriction justifications can enhance transparency and trust in the research.

Remark 5

Category: reproducibility

Location: Results

Issue: The lack of detailed data presentation and code results limits the reproducibility of the study findings.

Severity: high

Impact: Without clear data presentation and code results, replicating the study becomes challenging for other researchers.

Improvement Suggestions

Suggestion 1

Location: Results

Category: results

Focus: reproducibility

Original Text:

Neither the Nutrition Info, positive + Nutrition Info, high-in, nor positive + high-in labels led participants to make healthier purchases compared to the positive labels only.

Improved Version:

The Nutrition Info, positive + Nutrition Info, high-in, and positive + high-in labels did not result in participants making healthier purchases compared to the positive labels only.

Explanation:

Clarity and readability are improved by restructuring the sentence for better comprehension.

Suggestion 2

Location: Results

Category: results

Focus: reproducibility

Original Text:

The effects of the labels on overall healthfulness were not moderated by nutrition literacy, income, or educational attainment.

Improved Version:

The effects of the labels on overall healthfulness were consistent across different levels of nutrition literacy, income, and educational attainment.

Explanation:

Enhances clarity and provides a more concise description of the findings.

Suggestion 3

Location: Discussion

Category: discussion

Focus: reproducibility

Original Text:

In this study, Nutrition Info labels similar to FDA's proposed labels did not lead to healthier selections than positive labels.

Improved Version:

In this study, Nutrition Info labels, similar to the FDA's proposed labels, did not result in participants making healthier selections compared to positive labels.

Explanation:

Adding clarity and specificity by providing more context and details.

Suggestion 4

Location: Discussion

Category: discussion

Focus: reproducibility

Original Text:

Future research should explore how to design spectrum labels such that they are linked to federal regulations, such as federal guidelines for nutrient content or the FDA's definition of "healthy."

Improved Version:

Future research should investigate methods to align spectrum labels with federal regulations, such as incorporating federal guidelines for nutrient content or adhering to the FDA's definition of "healthy."

Explanation:

Enhances precision and clarity in conveying the recommendation for future research.

Suggestion 5

Location: Results

Category: results

Focus: reproducibility

Original Text:

The lack of detailed data presentation and code results limits the reproducibility of the study findings.

Improved Version:

Insufficient data presentation and code result details hinder the reproducibility of the study findings.

Explanation:

Enhances clarity and emphasizes the impact of the issue on reproducibility.

Detailed Feedback**Data Sharing Assessment**

The study lacks a clear statement on data availability, making it challenging for other researchers to access and verify the data used in the study.

Code Availability

There is no mention of a code repository for the implementation of the randomized trial, hindering the reproducibility of the study as other researchers cannot access the code.

Documentation Completeness

The documentation of the methodology for data collection and code implementation is insufficient, impacting the transparency and reproducibility of the research process.

Restrictions Justification

The study does not provide any justification for access restrictions to the data or code, limiting transparency and hindering the ability of other researchers to understand any limitations.

Reproducibility Support

The study lacks detailed data presentation and code results, which is crucial for reproducibility as it allows other researchers to replicate the study findings accurately.

R5 - Statistical Rigor

Score: 7/10

Summary

Overall, the study demonstrates good statistical rigor in terms of test selection, confidence interval reporting, and statistical power justification. However, improvements are needed in sample size justification, effect size reporting, and p-value interpretation to enhance the validity and practical significance of the findings.

Critical Remarks

Remark 1

Category: sample_size

Location: METHODS section

Issue: The sample size justification is based on detecting a small effect size (Cohen's $d=.15$), which may be considered too small for practical significance in this context.

Severity: medium

Impact: This may lead to underpowered analyses and limit the generalizability of the findings.

Remark 2

Category: p_value

Location: RESULTS section

Issue: The interpretation of p-values lacks context and does not consider the clinical or practical significance of the results.

Severity: medium

Impact: This may lead to misleading conclusions about the statistical significance of the findings.

Remark 3

Category: effect_size

Location: RESULTS section

Issue: The effect sizes reported are limited to average differential effects (ADEs) without considering standardized effect sizes like Cohen's d .

Severity: low

Impact: Including standardized effect sizes would provide a clearer understanding of the practical significance of the results.

Improvement Suggestions

Suggestion 1

Location: METHODS section

Category: sample_size

Focus: sample_size

Original Text:

Assuming a two-tailed critical alpha of 0.05, the target sample size of 5,610 (935 per arm) yielded 90% power to detect an effect of this size or larger.

Improved Version:

Consider reevaluating the sample size justification based on a larger effect size that is more practically significant in the context of the study.

Explanation:

This would ensure that the study is adequately powered to detect meaningful differences.

Suggestion 2

Location: RESULTS section

Category: p_value

Focus: p_value

Original Text:

The effects of the labels on overall healthfulness were not moderated by nutrition literacy, income, or educational attainment.

Improved Version:

The effects of the labels on overall healthfulness were not found to be significantly moderated by nutrition literacy, income, or educational attainment.

Explanation:

Adding the term 'significantly' clarifies the statistical interpretation.

Detailed Feedback

Test Selection

The study appropriately used ordinary least squares (OLS) regression to analyze the effects of front-of-package labels on healthfulness.

Assumption Verification

The assumption of linearity in OLS regression should be explicitly mentioned and verified in the methodology.

Sample Size Justification

The sample size justification could be improved by considering effect sizes that are more practically significant.

Multiple Comparison Handling

The study did not explicitly address multiple comparison issues; adjusting for multiple testing methods like Bonferroni correction could enhance the validity of the results.

Effect Size Reporting

While the study reported average differential effects (ADEs), including standardized effect sizes like Cohen's d would provide a more comprehensive understanding of the effect magnitudes.

Confidence Intervals

The study appropriately reported confidence intervals for the effect estimates, providing a range of plausible values for the population parameters.

P Value Interpretation

The interpretation of p-values could be enhanced by providing context on the clinical or practical significance of the results.

Statistical Power

The study adequately justified the sample size based on a specified effect size and desired power level.

Missing Data Handling

The study did not mention specific methods for handling missing data; addressing this issue explicitly would improve the transparency of the analysis.

Outlier Treatment

The study did not detail how outliers were identified or treated; providing information on outlier detection methods and their impact on the results would enhance the robustness of the analysis.

R6 - Technical Accuracy

Score: 8/10

Summary

The study demonstrates a strong technical foundation in analyzing the impact of front-of-package labeling systems on food purchases. However, there are areas for improvement in providing detailed derivations, ensuring consistency in presenting results, and enhancing the depth of discussion on the reasons behind the observed effects.

Critical Remarks

Remark 1

Category: mathematical_framework

Location: METHODS section

Issue: Lack of detailed explanation on how Ofcom scores were calculated from the United Kingdom's Nutrient Profiling Model.

Severity: medium

Impact: Affects the transparency and reproducibility of the study results.

Remark 2

Category: equations

Location: RESULTS section

Issue: Inconsistencies in the presentation of ADE values for different labeling systems.

Severity: low

Impact: May confuse readers and undermine the clarity of the results.

Remark 3

Category: documentation

Location: DISCUSSION section

Issue: Limited discussion on the potential reasons why certain labeling systems outperformed others.

Severity: medium

Impact: Reduces the depth of analysis and interpretation of the results.

Improvement Suggestions

Suggestion 1

Location: METHODS section

Category: mathematical_framework

Focus: terminology

Original Text:

We calculated Ofcom scores for all products in the store, converting scores to a 0-100 scale.

Improved Version:

Ofcom scores were derived from the United Kingdom's Nutrient Profiling Model and transformed to a 0-100 scale for consistency.

Explanation:

Clarifies the source of the scores and enhances reproducibility.

Suggestion 2

Location: RESULTS section

Category: equations

Focus: equations

Original Text:

The spectrum labels improved healthfulness by approximately 4% compared to the positive labels.

Improved Version:

The spectrum labels led to a 4% increase in healthfulness compared to the positive labels.

Explanation:

Enhances clarity and consistency in presenting results.

Detailed Feedback

Derivation Correctness

The study lacks detailed derivation explanations for the calculation of healthfulness scores, which could be crucial for understanding the methodology.

Algorithm Accuracy

The algorithm description for the randomized trial design is clear and well-structured, ensuring the reproducibility of the study.

Terminology Accuracy

The technical terminology used throughout the text is appropriate and aligns with the field of nutrition research.

Equation Clarity

Equations are presented in a clear and concise manner, aiding in the understanding of the results.

Content Completeness

The study covers a wide range of aspects related to front-of-package labeling systems, providing a comprehensive analysis.

Logical Consistency

The logical flow of the study is well-maintained, with clear connections between the objectives, methods, results, and conclusions.

Implementation Details

The implementation of the randomized trial in an online grocery store is well-described, allowing for replication of the study.

Edge Case Handling

The study lacks explicit discussion on how edge cases or outliers were handled in the analysis, which could impact the robustness of the findings.

Complexity Analysis

The complexity analysis of the labeling systems' effects on healthfulness is thorough, providing insights into the comparative effectiveness of different labels.

Technical Documentation

The technical documentation of the study design and procedures is detailed, enhancing the transparency and credibility of the research.

R7 - Consistency

Score: 7/10

Summary

The study demonstrates a strong alignment between the methods and results, with consistent testing of hypotheses and logical flow between sections. However, improvements are needed in terminology standardization, detailed presentation of data supporting conclusions, and better integration of the introduction with the study design. Enhancing the specificity of results and conclusions would further strengthen the coherence and consistency of the text.

Critical Remarks

Remark 1

Category: results_conclusions

Location: Results section

Issue: The spectrum labels are stated to be more effective in promoting healthier purchases, but the specific data supporting this claim is not clearly presented.

Severity: medium

Impact: Affects the credibility of the conclusion drawn from the results.

Remark 2

Category: logical_flow

Location: Introduction and Methods sections

Issue: The introduction discusses the FDA's proposed labels, but the Methods section lacks a detailed explanation of how these labels were implemented in the study.

Severity: medium

Impact: Inconsistency in providing a clear link between the theoretical framework and the study implementation.

Remark 3

Category: terminology

Location: Throughout the text

Issue: Inconsistent use of terminology such as 'healthfulness' and 'healthier purchases' without clear definitions or standardization.

Severity: medium

Impact: Confusion for readers and lack of clarity in conveying key concepts.

Improvement Suggestions

Suggestion 1

Location: Results section

Category: results

Focus: results_conclusions

Original Text:

The spectrum labels led participants to make healthier purchases compared to both the positive labels only and all other labeling systems.

Improved Version:

The spectrum labels led participants to make purchases with higher healthfulness scores compared to both the positive labels only and all other labeling systems.

Explanation:

Adding specificity by mentioning 'higher healthfulness scores' clarifies the comparison.

Suggestion 2

Location: Results section

Category: results

Focus: interpretation

Original Text:

The effects of the food labels on healthfulness did not vary by nutrition literacy, income, or educational attainment.

Improved Version:

The effects of the food labels on healthfulness were consistent across different levels of nutrition literacy, income, and educational attainment.

Explanation:

Enhancing clarity by specifying 'consistent across different levels' and restructuring the sentence for better readability.

Suggestion 3

Location: Introduction section

Category: literature

Focus: citations

Original Text:

Almost no studies have evaluated these Nutrition Info labels.

Improved Version:

Limited research has evaluated these Nutrition Info labels.

Explanation:

Using 'limited research' instead of 'almost no studies' provides a more accurate representation.

Detailed Feedback

Methods Results Alignment

The methods clearly outline the study design, but the results lack detailed statistical analysis supporting the conclusions.

Results Conclusions Alignment

While the results highlight the effectiveness of spectrum labels, the conclusions could be more nuanced and supported by specific data.

Logical Flow

The logical flow is generally coherent, but there is a need for better integration between the introduction and methods to ensure consistency in presenting the study design.

Terminology Consistency

The terminology related to healthfulness and label effectiveness needs standardization and clearer definitions throughout the text.

Hypothesis Testing

The study effectively tests the impact of different front-of-package labels on purchase healthfulness, aligning with the research objectives.

Interpretation Consistency

The interpretation of results could be strengthened by providing more detailed explanations of the observed effects and their implications.

Citation Consistency

The text maintains consistent citation practices, referencing prior studies and regulatory documents appropriately.

Figure Text Alignment

Figures are well-referenced and support the text, providing visual aids for better understanding.

Table Text Alignment

Tables are effectively used to present data, but some tables could benefit from clearer explanations and headings for better alignment with the text.

Supplementary Consistency

The supplementary material complements the main text, providing additional details and supporting information.

Writing Agents (W1-W8)

W1 - Language and Style

Score: 7/10

Summary

Overall, the text demonstrates good language and style, with minor issues in grammar, spelling, and punctuation. Improvements in sentence structure, verb tense consistency, and subject-verb agreement could enhance readability. Attention to detail in spelling and punctuation will further refine the academic tone of the text.

Critical Remarks

Remark 1

Category: grammar

Location: Abstract

Issue: The sentence 'The spectrum labels led participants to make healthier purchases compared to both the positive labels only' lacks a verb, making it incomplete.

Severity: medium

Impact: Affects clarity and completeness of the statement.

Remark 2

Category: spelling

Location: Introduction

Issue: The word 'guidlines' should be corrected to 'guidelines'.

Severity: low

Impact: Minor spelling error.

Remark 3

Category: punctuation

Location: Methodology

Issue: The sentence 'We assessed the effects of 6 front-of-package labeling systems' lacks a comma after 'systems'.

Severity: low

Impact: Minor punctuation error.

Improvement Suggestions

Suggestion 1

Location: Abstract

Category: abstract

Focus: grammar

Original Text:

The spectrum labels led participants to make healthier purchases compared to both the positive labels only.

Improved Version:

The spectrum labels led participants to make healthier purchases compared to the positive labels only.

Explanation:

Adding the verb 'led' makes the sentence grammatically correct and complete.

Suggestion 2

Location: Introduction

Category: introduction

Focus: spelling

Original Text:

guidlines

Improved Version:

guidelines

Explanation:

Correcting the spelling of 'guidlines' to 'guidelines' is necessary for accuracy.

Suggestion 3

Location: Methodology

Category: methodology

Focus: punctuation

Original Text:

We assessed the effects of 6 front-of-package labeling systems

Improved Version:

We assessed the effects of 6 front-of-package labeling systems.

Explanation:

Adding a comma after 'systems' improves the clarity and flow of the sentence.

Detailed Feedback

Grammar Correctness

The text generally demonstrates good grammar, but occasional errors in sentence structure and verb tense consistency are present.

Spelling Accuracy

Spelling is mostly accurate, with a few minor errors that need attention.

Punctuation Usage

Punctuation is adequate, but some sentences could benefit from additional commas for clarity.

Sentence Structure

The overall sentence structure is clear, but some sentences could be improved for better readability.

Verb Tense Consistency

Verb tense consistency is generally maintained, but a few instances of inconsistency exist.

Subject Verb Agreement

Subject-verb agreement is mostly correct, but a few instances require attention.

Article Usage

Article usage is appropriate throughout the text.

Preposition Usage

Prepositions are used correctly, contributing to the overall coherence of the text.

Conjunction Usage

Conjunctions are appropriately used to connect ideas and maintain coherence.

Academic Conventions

The text adheres well to academic writing conventions, maintaining a formal and scholarly tone.

W2 - Narrative and Structure

Score: 7/10

Summary

The study demonstrates a logical progression of ideas, but improvements in section transitions, paragraph organization, and conclusion alignment are needed to enhance narrative coherence. Strengthening topic sentences, supporting evidence integration, and hypothesis tracking would further improve the logical flow and reader engagement.

Additionally, clearer connections between sections, improved visual element integration, and enhanced reader engagement strategies would elevate the overall narrative structure.

Critical Remarks

Remark 1

Category: paragraph_organization

Location: Introduction

Issue: The introduction lacks a clear roadmap of the study, making it challenging for readers to anticipate the content.

Severity: medium

Impact: Readers may struggle to grasp the study's purpose and relevance.

Remark 2

Category: evidence_integration

Location: Results

Issue: The presentation of results lacks clear connections to the research question and hypothesis.

Severity: high

Impact: Readers may find it difficult to understand how the results address the study objectives.

Remark 3

Category: conclusion_alignment

Location: Conclusion

Issue: The conclusion does not directly tie back to the introduction, missing an opportunity to reinforce the study's significance.

Severity: medium

Impact: Readers may feel disconnected from the study's progression and outcomes.

Improvement Suggestions

Suggestion 1

Location: Introduction

Category: introduction

Focus: narrative_coherence

Original Text:

The introduction lacks a clear roadmap of the study.

Improved Version:

The introduction should provide a concise overview of the study objectives, research questions, and the importance of the study.

Explanation:

This improvement will help orient readers and set expectations for the content that follows.

Suggestion 2

Location: Results

Category: results

Focus: logical_progression

Original Text:

Results lack clear connections to the research question.

Improved Version:

Results should be explicitly linked to the research question and hypothesis to demonstrate how they address the study objectives.

Explanation:

This enhancement will enhance the coherence and relevance of the results section.

Suggestion 3

Location: Conclusion

Category: conclusion

Focus: conclusion_alignment

Original Text:

Conclusion does not directly tie back to the introduction.

Improved Version:

The conclusion should reiterate key findings and their alignment with the study's initial objectives outlined in the introduction.

Explanation:

This adjustment will strengthen the narrative flow and reinforce the study's significance.

Detailed Feedback

Narrative Coherence

The text demonstrates a logical flow of ideas, but improvements are needed in connecting sections more explicitly to enhance overall coherence.

Logical Progression

The study progresses logically from introduction to conclusion, but some sections could benefit from clearer transitions and connections to maintain a smooth flow of information.

Section Transitions

While most sections transition adequately, enhancing the clarity of transitions between the introduction, literature review, and methodology could improve the overall structural organization.

Paragraph Organization

Paragraphs are generally well-structured, but ensuring each paragraph focuses on a single main idea and aligns with the overall section theme would enhance readability and comprehension.

Topic Sentence Effectiveness

Topic sentences effectively introduce paragraph content, but strengthening their alignment with the main thesis of each section could improve the overall coherence.

Supporting Evidence Integration

The integration of supporting evidence is generally effective, but ensuring a clear link between evidence and the main argument in each section would enhance the persuasive impact.

Conclusion Alignment

The conclusion aligns with the study's findings, but reinforcing connections to the introduction and restating the main contributions would enhance the narrative closure.

Hypothesis Tracking

The study effectively tracks the research question throughout, but explicitly linking results to the initial hypothesis in each section would strengthen the hypothesis tracking.

Visual Element Integration

The inclusion of figures and tables enhances the visual narrative, but providing more detailed explanations and interpretations alongside visual elements could improve reader engagement.

Reader Engagement

The text engages readers with relevant information, but incorporating more interactive elements, such as questions for reflection or interactive figures, could further enhance reader engagement.

W3 - Clarity and Conciseness

Score: 6/10

Summary

The text exhibits moderate clarity and conciseness issues, primarily related to jargon usage, wordiness, and redundancy. Simplifying language, reducing complexity, and enhancing sentence structures would significantly improve readability and comprehension.

Critical Remarks

Remark 1

Category: jargon

Location: Abstract

Issue: The use of technical terms like 'Ofcom Nutrient Profiling Model scores' may be unclear to a general audience.

Severity: medium

Impact: This can hinder understanding of the study's objectives.

Remark 2

Category: wordiness

Location: Introduction

Issue: The paragraph discussing the causes of poor diet quality is verbose and could be more concise.

Severity: low

Impact: Reduces the clarity and makes the text less engaging.

Remark 3

Category: redundancy

Location: Results

Issue: Repetition of the phrase 'positive labels' in multiple sentences.

Severity: low

Impact: Creates unnecessary redundancy and affects the flow of information.

Improvement Suggestions

Suggestion 1

Location: Abstract

Category: abstract

Focus: language_simplicity

Original Text:

To test whether 'Nutrition Info' front-of-package food labels proposed by the US Food and Drug Administration (FDA) increase purchase healthfulness compared to other existing or proposed food labeling systems.

Improved Version:

To evaluate if the 'Nutrition Info' labels proposed by the FDA enhance the healthfulness of purchases compared to other labeling systems.

Explanation:

Simplified the sentence by removing unnecessary details and jargon.

Suggestion 2

Location: Introduction

Category: introduction

Focus: wordiness

Original Text:

Poor diet quality is responsible for approximately 500,000 deaths and more than \$50 billion in healthcare costs in the US every year.

Improved Version:

Poor diet quality causes 500,000 deaths and over \$50 billion in healthcare costs annually in the US.

Explanation:

Condensed the information for better readability and conciseness.

Suggestion 3

Location: Discussion

Category: discussion

Focus: language_simplicity

Original Text:

In this randomized trial in a large sample of US adults, 'spectrum' food labels that rated all foods from least to most healthy led participants to select modestly healthier foods and beverages compared to the status quo of displaying only positive labels on healthier foods.

Improved Version:

This study found that 'spectrum' food labels, rating foods from least to most healthy, resulted in slightly healthier food choices compared to the current practice of using positive labels on healthier foods.

Explanation:

Clarified the sentence structure and simplified the language.

Detailed Feedback

Language Simplicity

The text uses technical terms and complex language that may be challenging for a general audience to understand. Simplifying the vocabulary and sentence structures would enhance clarity.

Jargon Usage

There is a significant amount of jargon related to food labeling systems, nutrient profiling, and regulatory terms. Explaining these terms in simpler language or providing definitions would improve comprehension.

Wordiness

Several paragraphs contain unnecessary wordiness and could be more concise without losing important information. Streamlining the text would make it more engaging and easier to follow.

Sentence Length

The text contains long sentences that could be broken down into shorter, more digestible segments. Shorter sentences improve readability and help convey information more effectively.

Paragraph Length

Some paragraphs are lengthy and cover multiple ideas, leading to information overload. Breaking down paragraphs into smaller, focused sections would enhance readability and understanding.

Active Passive Voice

The text predominantly uses passive voice, which can make sentences less direct and engaging. Incorporating more active voice would improve clarity and make the text more dynamic.

Redundancy

There are instances of redundant phrases and repetitive information that could be streamlined to avoid unnecessary repetition and maintain a smooth flow of information.

Ambiguity

Certain sections contain ambiguous terms or vague descriptions that could be clarified for better understanding. Providing specific examples or explanations would reduce ambiguity.

Readability

Overall, the text's readability could be improved by simplifying language, breaking down complex sentences, and organizing information into more digestible chunks.

Information Density

The text contains dense information, which may overwhelm readers. Balancing the amount of information presented and ensuring it is structured logically would enhance information density.

W4 - Terminology Consistency

Score: 6/10

Summary

The text exhibits moderate terminology consistency, with notable areas for improvement in term usage, notation, and acronym definition. Enhancing these aspects will elevate the overall clarity and professionalism of the document.

Critical Remarks

Remark 1

Category: term_usage

Location: Abstract

Issue: Inconsistency in using 'healthfulness' and 'healthier purchases' interchangeably.

Severity: medium

Impact: Confusion in conveying the precise meaning of the outcomes.

Remark 2

Category: notation

Location: Equations

Issue: Inconsistent notation style in equations.

Severity: low

Impact: Affects the clarity and professional presentation of mathematical expressions.

Remark 3

Category: acronyms

Location: Introduction

Issue: Acronyms like 'US' and 'FDA' are used without prior definition.

Severity: medium

Impact: Readers unfamiliar with the acronyms may face comprehension challenges.

Improvement Suggestions

Suggestion 1

Location: Abstract

Category: abstract

Focus: term_usage

Original Text:

healthfulness of purchases

Improved Version:

healthfulness of food and beverage selections

Explanation:

Clarity in specifying the type of selections being analyzed.

Suggestion 2

Location: Equations

Category: equations

Focus: notation

Original Text:

ADE=-.04

Improved Version:

Average Differential Effect (ADE) of -0.04

Explanation:

Enhances the formal presentation of results.

Suggestion 3

Location: Introduction

Category: introduction

Focus: acronyms

Original Text:

US adults

Improved Version:

adults in the United States (US)

Explanation:

Provides clarity and avoids ambiguity.

Detailed Feedback

Term Usage Consistency

The text demonstrates moderate consistency in term usage, but there are instances where synonyms are used interchangeably, leading to potential confusion.

Notation Consistency

Notation consistency is generally maintained, but there are sporadic instances of inconsistency in mathematical expressions.

Acronym Usage

Acronyms are used without prior definition in some sections, impacting reader understanding.

Variable Naming Consistency

Variable naming is consistent throughout the text, enhancing readability and comprehension.

Unit Notation Consistency

Unit notation is consistently applied, contributing to the clarity of measurements and results.

Abbreviation Consistency

Abbreviations are used consistently, aiding in brevity and readability.

Technical Term Consistency

Technical terms are appropriately used and defined, maintaining clarity and precision.

Field Terminology

The field-specific terminology is well-incorporated, aligning with the subject matter and enhancing domain-specific understanding.

Cross Reference Consistency

Cross-references are consistent and aid in connecting related information throughout the text.

Definition Consistency

Definitions are clear and consistent, providing a solid foundation for understanding key concepts.

W5 - Inclusive Language

Score: 7/10

Summary

The text demonstrates a good level of inclusivity in terms of gender-neutral language and age-appropriate terminology. However, there are opportunities to enhance cultural sensitivity, socioeconomic representation, and disability inclusion. Improvements in professional title usage and geographic inclusivity can further enhance the study's inclusivity and relevance.

Critical Remarks

Remark 1

Category: gender_neutrality

Location: Authors section

Issue: Gender-specific titles like 'Mr.' or 'Ms.' are used for some authors.

Severity: medium

Impact: Excludes non-binary or gender-diverse individuals from the authorship representation.

Remark 2

Category: cultural_sensitivity

Location: Introduction section

Issue: The text focuses primarily on US data and policies, neglecting global perspectives on nutrition labeling.

Severity: medium

Impact: Lacks inclusivity towards diverse cultural contexts and practices in nutrition labeling.

Remark 3

Category: socioeconomic_sensitivity

Location: Results section

Issue: Income brackets are limited and may not fully capture the socioeconomic diversity of participants.

Severity: low

Impact: May oversimplify the socioeconomic representation in the study.

Improvement Suggestions

Suggestion 1

Location: Authors section

Category: professional_titles

Focus: gender_neutrality

Original Text:

Authors: Anna H. Grummon,^{1,2} Kevin O'Sullivan,³ Joshua Petimar,⁴ Cristina J.Y. Lee,¹ Amanda B. Zeitlin,¹ Lauren P. Cleveland,⁵ Caroline Collis,⁵ Aviva A. Musicus,^{6, 7} Verena Tiefenbeck,⁸ Elgar Fleisch,³ and Jason P. Block⁴

Improved Version:

Authors: Anna H. Grummon, Kevin O'Sullivan, Joshua Petimar, Cristina J.Y. Lee, Amanda B. Zeitlin, Lauren P. Cleveland, Caroline Collis, Aviva A. Musicus, Verena Tiefenbeck, and Elgar Fleisch

Explanation:

Removing gender-specific titles provides a more inclusive representation of the authors.

Suggestion 2

Location: Introduction section

Category: geographic_inclusivity

Focus: cultural_sensitivity

Original Text:

The US requires most packaged foods to display the Nutrition Facts Label, but this label is displayed on the back or side of product packaging...

Improved Version:

The US requires most packaged foods to display the Nutrition Facts Label, but this label is typically located on the back or side of product packaging...

Explanation:

Adding 'typically' acknowledges variations in label placement, enhancing accuracy.

Suggestion 3

Location: Participant characteristics table

Category: gender_neutrality

Focus: identity_language

Original Text:

The average age of participants was 40.3 years (SD=12.6); 60% identified as female...

Improved Version:

The average age of participants was 40.3 years (SD=12.6); 60% identified as women...

Explanation:

Using 'women' instead of 'female' is more person-centered and respectful.

Detailed Feedback**Gender Neutral Language**

The text generally uses gender-neutral language in participant descriptions, which is positive for inclusivity.

Cultural Sensitivity

The study focuses heavily on US contexts, potentially limiting the cultural sensitivity and applicability to diverse global settings.

Age Appropriate Terminology

The use of age-appropriate terminology is consistent and respectful throughout the text.

Disability Inclusive Language

There are no explicit references to disability-related language or considerations in the text.

Socioeconomic Sensitivity

The study provides limited income brackets, which may not fully capture the socioeconomic diversity of participants.

Geographic Inclusivity

The study predominantly discusses US policies and data, lacking a broader global perspective on nutrition labeling.

Professional Title Usage

The use of professional titles for authors is gender-specific in some cases, impacting gender neutrality.

Stereotypes

The text generally avoids stereotypes and presents information objectively.

Identity Language

The text uses person-first language in participant descriptions, enhancing respectful representation.

Historical Context

The study acknowledges the historical context of FDA regulations and global labeling practices, providing relevant background information.

W6 - Citation Formatting

Score: 5/10

Summary

The citation formatting in the text is generally acceptable, with minor issues in author name consistency, missing DOI information, and inconsistent table formatting. Improvements are needed in standardizing author names, including DOIs for all references, and providing clear headings for tables to enhance readability and professionalism.

Critical Remarks

Remark 1

Category: author_format

Location: Authors section

Issue: Inconsistent author name formatting with some initials and some full names.

Severity: medium

Impact: Affects the professionalism and consistency of the citation.

Remark 2

Category: doi_format

Location: References section

Issue: Missing DOI information for some references.

Severity: low

Impact: Lack of DOI reduces the accessibility and credibility of the sources.

Remark 3

Category: volume_format

Location: Results section

Issue: Inconsistent presentation of data in tables without clear headings.

Severity: medium

Impact: Confusion for readers in understanding the results.

Improvement Suggestions

Suggestion 1

Location: Authors section

Category: author_format

Focus: author_format

Original Text:

Anna H. Grummon,^{1,2} Kevin O'Sullivan,³ Joshua Petimar,⁴ Cristina J.Y. Lee,¹ Amanda B. Zeitlin,¹ Lauren P. Cleveland,⁵ Caroline Collis,⁵ Aviva A. Musicus,^{6, 7} Verena Tiefenbeck,⁸ Elgar Fleisch,³ and Jason P. Block⁴

Improved Version:

Grummon AH, O'Sullivan K, Petimar J, Lee CJY, Zeitlin AB, Cleveland LP, Collis C, Musicus AA, Tiefenbeck V, Fleisch E, Block JP

Explanation:

Consistent formatting with last name followed by initials for all authors.

Suggestion 2

Location: References section

Category: doi_format

Focus: doi_format

Original Text:

doi:10.1016/S0140-6736(19)30041-8

Improved Version:

DOI: 10.1016/S0140-6736(19)30041-8

Explanation:

Standardize the DOI format with a clear label for easy identification.

Suggestion 3

Location: Results section

Category: volume_format

Focus: volume_format

Original Text:

Guiding Stars score (-22 to 7)

Improved Version:

Guiding Stars score (Range: -22 to 7)

Explanation:

Clarify the range of the Guiding Stars score for better understanding.

Detailed Feedback

In Text Citation Format

In-text citations are consistently formatted with author names and publication years.

Reference List Format

The reference list follows a standard format with author names, publication dates, titles, and source information.

Citation Style Consistency

Overall, the citation style is mostly consistent with minor inconsistencies in author name formatting.

Reference Completeness

References are complete with necessary information, but some lack DOI details.

Doi Url Formatting

DOIs and URLs are present but not consistently formatted with a clear label for each.

Author Name Formatting

Author names are mostly formatted correctly but need consistency in using initials or full names.

Publication Date Formatting

Publication dates are presented in a clear and consistent format throughout the text.

Journal Name Formatting

Journal names are correctly formatted and consistent in the references.

Volume Issue Page Formatting

Volume, issue, and page numbers are provided where applicable, but some tables lack clear headings.

Cross Reference Accuracy

Cross-references are accurate and relevant within the text.

W7 - Target Audience Alignment

Score: 0/10

Summary

The study demonstrates a strong technical depth and effective use of field-specific terminology. However, improvements in writing formality, visual integration, and reference depth could enhance audience alignment. Enhancing the methodology detail, results presentation, discussion depth, and conclusion format would further strengthen the study's impact.

Critical Remarks

Remark 1

Category: technical_depth

Location: METHODS section

Issue: Lack of detailed explanation on statistical analysis techniques used in the study.

Severity: medium

Impact: May hinder understanding for readers not familiar with advanced statistical methods.

Remark 2

Category: terminology

Location: RESULTS section

Issue: Overuse of technical jargon without clear explanations, making it hard for non-experts to follow.

Severity: high

Impact: Could alienate a broader audience and limit comprehension.

Remark 3

Category: visuals

Location: Figure 2

Issue: Lack of detailed explanation in the CONSORT flow diagram.

Severity: medium

Impact: Readers may struggle to interpret the flow of participants through the study.

Improvement Suggestions

Suggestion 1

Location: Abstract

Category: abstract

Focus: conciseness

Original Text:

We conducted a randomized trial in October-November 2024 (n=5,636 US adults).

Improved Version:

In October-November 2024, we conducted a randomized trial involving 5,636 US adults to assess...

Explanation:

Adding 'involving' provides clarity on the purpose of the trial upfront.

Suggestion 2

Location: RESULTS section

Category: results

Focus: technical_depth

Original Text:

The effects of the labels on overall healthfulness were not moderated by nutrition literacy, income, or educational attainment.

Improved Version:

The effects of the labels on overall healthfulness were not influenced by participants' nutrition literacy, income levels, or educational attainment.

Explanation:

Expanding 'influenced by' helps clarify the lack of impact on different participant characteristics.

Detailed Feedback

Technical Depth

The study provides a detailed methodology description, including participant recruitment and measures. However, more explanation on the statistical analysis methods used could enhance the technical depth for readers.

Terminology Usage

The study employs field-specific terminology effectively, but some terms are not adequately explained, impacting reader comprehension.

Writing Formality

The writing style is formal and appropriate for an academic audience. However, some sections could benefit from simplification for broader understanding.

Section Organization

The study is well-organized with clear section headings. However, improving the flow between sections could enhance coherence.

Visual Integration

Visual elements like figures and tables are used effectively. However, some visuals lack detailed explanations, affecting audience interpretation.

Reference Style

The study includes a comprehensive list of references, enhancing the depth of research. However, more in-text citations could strengthen the support for claims.

Methodology Detail

The methodology description is thorough, outlining participant recruitment, setting, and procedures. Adding more detail on statistical analysis techniques could further enhance the methodology section.

Results Presentation

Results are presented clearly with statistical data. However, simplifying technical terms and providing more context could improve audience understanding.

Discussion Depth

The discussion provides insightful analysis of the results. Expanding on the implications of the findings and discussing potential limitations could enhance the depth of the discussion.

Conclusion Format

The conclusion summarizes the key findings effectively. Adding recommendations for future research and policy implications could enhance the conclusion format.

W8 - Visual Presentation

Score: 6/10

Summary

Overall, the manuscript could benefit from enhancements in figure quality, table formatting, and caption completeness to improve visual presentation quality and effectiveness.

Critical Remarks

Remark 1

Category: Figure quality and clarity

Location: Figure 1

Issue: The labels in Figure 1 could be more visually appealing and distinct to enhance clarity and effectiveness.

Severity: medium

Impact: May lead to confusion or difficulty in distinguishing between different labels.

Remark 2

Category: Table formatting and readability

Location: Table 1

Issue: The table layout could be improved for better readability and organization of participant characteristics.

Severity: medium

Impact: May hinder quick comprehension of participant demographics.

Remark 3

Category: Caption completeness

Location: Table 2

Issue: The table lacks detailed captions explaining the outcomes and ADE values, affecting clarity.

Severity: low

Impact: Readers may struggle to understand the significance of the data presented.

Improvement Suggestions

Suggestion 1

Location: Figure 1

Category: Figure

Focus: Caption clarity

Original Text:

Figure 1: Examples of labels tested in the trial

Improved Version:

Figure 1: Visual representation of different front-of-package labels tested in the trial with clear and distinct designs.

Explanation:

Enhances clarity and effectiveness by specifying the purpose and content of the figure.

Suggestion 2

Location: Table 1

Category: Table

Focus: Title clarity

Original Text:

Table 1. Participant characteristics, n=5,636 US adults

Improved Version:

Table 1. Demographic Characteristics of Participants (n=5,636)

Explanation:

Provides a more focused and descriptive title for better understanding.

Suggestion 3

Location: Table 2

Category: Table

Focus: Title clarity

Original Text:

Table 2. Impact of front-of-package labeling systems on food and beverage selections, n=5,636 US adults

Improved Version:

Table 2. Effect of Front-of-Package Labels on Food Selections (n=5,636)

Explanation:

Clarifies the purpose of the table and emphasizes the key information.

Detailed Feedback**Figure Quality**

The visual elements could be more distinct and visually appealing.

Table Formatting

Tables could benefit from improved layout and organization for better readability.

Visual Placement

Visuals should be strategically placed to enhance understanding and flow.

Caption Completeness

Captions should be detailed and informative to provide context.

Color Scheme

Appropriate color schemes should be used to aid in differentiation and interpretation.

Data Visualization

Data visualization should be optimized for clear representation and easy interpretation.

Visual Hierarchy

Visual hierarchy should be established to guide the reader's attention effectively.

Accessibility

Ensure color contrast and readability for improved accessibility.

Visual Consistency

Maintain consistency in visual style and formatting across all elements.

Text Integration

Visuals should seamlessly integrate with the text to support and enhance understanding.